Hybrid Cognitive-Fluid Dynamics for Language Production: A Wasserstein Gradient Flow Perspective

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Abstract

We introduce a family of reaction-diffusion models for the four-species Levelt–Baddeley account of language production, formulated as a Wasserstein gradient flow on the space of activation distributions. Using a multi-scale loop that couples Langevin particle dynamics with an implicit JKO step, we show that (i) the flow is numerically tractable at scale on commodity Apple Silicon (5 seeds $\times$ 10k particles $\times$ 100 slow steps in under four minutes with a fast projected-gradient path and under thirteen minutes with the full entropic-Wasserstein JKO path, on CPU); (ii) the fast path induces emergent species-level specialization when the attraction-to-species-regions is strong enough, with a measured entropy gap of 2.04 bits below the maximum at the best point of a 72-configuration sweep (51% below max), versus ~ 0 bits under the default flat potentials; and (iii) a compact neural surrogate distilled from the flow meets a sub-10 ms p50 streaming-inference budget on Apple Silicon with a measured p50 of 0.04 ms. The fast and full paths do not agree numerically (KL $\approx$ 0.26 nats at the final step), so results obtained with the fast path describe that scheme. We additionally report the measured ε-dependence of the entropic proximal step and a distributional continual-learning proxy that surfaces a stability/plasticity trade-off. The companion repository contains the full implementation, tests, and reproducibility scripts.

1 Introduction

Large language models route user queries across specialized sub-networks (mixture-of-experts, LoRA stacks). Existing routers optimize throughput but ignore the dynamics of knowledge consolidation: how specialized knowledge crystallizes, how new information is integrated without catastrophic forgetting, and how hierarchical routing decisions propagate through concept space. Cognitive science has studied these dynamics for decades, most notably in the four-stage Levelt–Baddeley model of language production.

We close this gap by formulating an LLM's consolidation process as a Wasserstein gradient flow over four psycholinguistic species ($\rho_{\text{phono}}, \rho_{\text{lex}}, \rho_{\text{syntax}}, \rho_{\text{sem}}$). Our contributions:

- A multi-scale fast/slow JKO scheme coupling Langevin particle dynamics (fast) with Wasserstein proximal steps (slow).
- A three-track implementation — performance-oriented, paper-grade, and production-deployable — released at `https://github.com/electron-rare/kiki-flow-research`.
- A quantitative characterization of the final-state entropy and peak amplitude on five independent seeds, surfacing the honest negative finding that the current functional stays near uniform.

- A neural surrogate distilled from the multi-scale trajectories, deployable in a production LLM router under 10 ms p50 on Apple Silicon (measured p50 = 0.04 ms).

2 Background and related work

Psycholinguistics of language production. Levelt's *Speaking* [16] formalizes language production as a four-stage cascade: a semantic conceptualizer produces a preverbal message; lexical access selects lemmas and retrieves lexemes; syntactic framing arranges the lexemes into a phrase structure; and phonological encoding yields the articulatory plan. Each stage has its own representation and feeds forward, with limited backwards feedback for error repair. Baddeley's working-memory model [2] localizes the phonological loop as a finite-capacity sub-vocal rehearsal buffer and adds an episodic buffer binding lexical, semantic, and temporal information. We take the four species $(\rho_{\text{phono}}, \rho_{\text{lex}}, \rho_{\text{syntax}}, \rho_{\text{sem}})$ to correspond to activation densities at each of Levelt's four stages, with coupling strengths derived from the literature on spreading activation and interactive-activation models. Direct evidence that LLMs represent such stage-specific factors is sparse; one example is [14], who find that the phonological condition for English *a/an* selection is encoded along a single linear direction that causally drives article choice, with the model forecasting the phonology of the upcoming word. We therefore treat the four-species decomposition as a modelling hypothesis, not as an established description of LLM internals.

Wasserstein gradient flows and the JKO scheme. Jordan, Kinderlehrer, and Otto [13] introduced the JKO scheme: an implicit-time discretization of the Fokker–Planck equation as the gradient flow of free energy in the Wasserstein-2 metric. Each step minimizes a free-energy functional regularized by the W_2 distance to the previous density. The scheme underpins both recent theoretical work on mean-field neural networks [18] (where training dynamics is cast as a Wasserstein flow over parameter distributions) and practical denoising/diffusion generative models (where reverse-time flows produce samples). For numerical tractability we replace exact optimal transport with entropic Sinkhorn, following the entropic JKO scheme of Peyré [20], and accept a quantifiable bias [10] in exchange for computational feasibility. The debiased Sinkhorn divergence approximates the squared Wasserstein-2 distance as $\varepsilon \to 0$, and [19] prove that the associated Sinkhorn Riemannian metric converges as $\varepsilon \to 0$ for finitely supported measures, the particle setting used here. With regularization $\varepsilon = \alpha\tau$, the entropic JKO scheme converges as $\tau \to 0$ to the target PDE plus a linear diffusion term $\frac{\alpha}{2}\Delta\rho$ [4]; a convergence rate is available under convexity assumptions as $\alpha, \tau \to 0$ [3]. Recent work casts a broad class of generative models as parametric JKO discretizations of Wasserstein gradient flows [6], and W-Flow [12] trains a static one-step generator to compress a Wasserstein gradient flow whose energy is a Sinkhorn divergence; both are close in spirit to Track 3, which distills Track 2 trajectories into a fast surrogate. Within an LLM pipeline, JKO-RAG [21] frames retrieval reranking as a W_2 free-energy JKO step and reports that the W_2 geometry damps the mass transport induced by query paraphrase, an independent use of a JKO step as a stability regularizer for LLM routing-type decisions. For the cross-diffusion systems that motivate the Turing coupling term, [9] review the existence theory of Wasserstein gradient flows built from JKO schemes. In the searches behind this draft we found no Wasserstein or JKO treatment of the dynamics of language-model representations during consolidation; we report this as the absence of a hit in a literature search, not as a proof of novelty.

Continual learning and consolidation. Elastic Weight Consolidation [15] regularizes network plasticity toward weights important for previous tasks, using the Fisher information as a penalty.

Our consolidation step plays an analogous role but operates in the distributional space of activation species rather than on individual weights: anti-forgetting is encoded by the W_2 proximal term of the JKO scheme, which prevents the new density from drifting too far from its predecessor. This shift from parameter-space to distribution-space regularization integrates naturally with mixture-of-experts and LoRA-stack architectures where the routing decision is itself a distributional choice. Sleep-inspired replay offers complementary consolidation baselines: [5] show that an unsupervised sleep-like replay phase applied after several sequentially trained tasks partially restores performance on all of them, whereas [22] confine replay to the degrees of freedom left unused by the current input and so dispense with a separate offline phase. Our nightly offline consolidation (Track 1) is closer to the former; neither is compared against in this paper.

Neuromorphic and mean-field inference. The reaction-diffusion coupling we inherit from Levelt–Baddeley is structurally similar to FitzHugh–Nagumo and other excitable-media models that have been used to study neural population dynamics. Turing's original instability analysis describes how cross-diffusion can spontaneously produce spatial patterns; we include a Turing term for completeness but find (Section 5) that it remains below the threshold for visible patterning at our scale and hyperparameters.

3 Model

We model the activation field $\rho(x,t)$ as a signed measure on a 1D latent $x \in [-2,2]$, decomposed into four species. At each consolidation step τ, we solve

$$\rho^{\tau+1} = \arg\min_{\rho} \Big\{ F[\rho] + \tfrac{1}{2h_\tau} \sum_i W_2^2(\rho_i, \rho_i^\tau) \Big\}, \tag{1}$$

where F combines a species-specific potential, a KL prior, a symmetric reaction coupling, and a Turing cross-diffusion term:

$$F[\rho] = \underbrace{\sum_i \int \rho_i V_i + \sum_i \mathrm{KL}(\rho_i \| \pi_i)}_{F_{\mathrm{sep}}(\rho)} + \tfrac{1}{2} \sum_{i,j} J_{ij}^{\mathrm{sym}} \int \rho_i \rho_j + \lambda_T \mathcal{T}[\rho], \tag{2}$$

where $J = J^{\mathrm{sym}} + J^{\mathrm{asym}}$ is the Levelt–Baddeley coupling matrix decomposed canonically into $J^{\mathrm{sym}} = \frac{1}{2}(J + J^\top)$ and $J^{\mathrm{asym}} = \frac{1}{2}(J - J^\top)$. On our default instantiation (Section 3.1) the asymmetry is structural: $\|J^{\mathrm{asym}}\|_F / \|J\|_F = 0.314$ % from kiki_flow_core/species/data/dell_baddeley_coupling.yaml. Only J^{sym} enters (2) because the antisymmetric bilinear form $\sum_{i,j} J_{ij}^{\mathrm{asym}} \int \rho_i \rho_j$ vanishes identically against the symmetric outer product. The antisymmetric component instead drives a non-conservative field

$$b_i^{\mathrm{asym}}(\rho) \;=\; \sum_j J_{ij}^{\mathrm{asym}} \, \rho_j, \tag{3}$$

which has no scalar-potential representation. Our dynamics therefore factor into a JKO step on F composed with an operator-splitting step on $b = b^{\mathrm{asym}} + b^{\mathrm{phono}}$, where b^{phono} is the Baddeley phonological-loop drift formalized in Section 3.3.

3.1 Coupling topologies: Dell and Levelt

We study two instantiations of J, each loaded from a YAML configuration so that the four-species dynamics can be compared across psycholinguistic architectures % from kiki_flow_core/species/data/{dell,levelt}_bado

Dell topology. Direct semantic-to-lexical drive $J_{\text{lex}\leftarrow\text{sem}} = 0.60$ that bypasses syntax, plus a weak $J_{\text{syn}\leftarrow\text{sem}} = 0.10$, giving the interactive lexical network of Dell [8].

Levelt topology. Canonical production chain sem $\rightarrow$ syn $\rightarrow$ lex $\rightarrow$ phono with balanced weights $J_{\text{syn}\leftarrow\text{sem}} = 0.45$, $J_{\text{lex}\leftarrow\text{syn}} = 0.40$, $J_{\text{phono}\leftarrow\text{lex}} = 0.50$, and the direct $J_{\text{lex}\leftarrow\text{sem}}$ reduced to 0.10. This matches the staged production architecture of Levelt [17].

Both topologies share the Baddeley phonological-loop backedge $J_{\text{lex}\leftarrow\text{phono}} = 0.30$ and the same diagonal self-consolidation weights. Frobenius norms are $\|J_{\text{Dell}}\|_F = 1.323$ and $\|J_{\text{Levelt}}\|_F = 1.247$; the asymmetry ratios $\|J^{\text{asym}}\|_F/\|J\|_F$ are 0.314 and 0.330 respectively, isolating topology as the primary difference between the two conditions.

3.2 Empirical λ-monotonicity

Following [7], the multi-species flow is λ-monotone at rate $\lambda > 0$ if, for any two trajectories $\rho(t), \rho'(t)$ from distinct initial conditions,

$$\frac{d}{dt}\sum_i W_2^2(\rho_i(t), \rho_i'(t)) \;\le\; -2\lambda \sum_i W_2^2(\rho_i(t), \rho_i'(t)). \tag{4}$$

We verify (4) numerically (Section 5.5; script `experiments/check_lambda_monotonicity.py`) on three independent seed pairs with $h = 2 \times 10^{-3}$ and $n_{\text{inner}} = 1$, using the Sinkhorn divergence [10] to cancel the entropic bias floor inherent to regularized W_2^2. We obtain

$$\hat{\lambda} \;=\; 67.79 \pm 3.78, \qquad \text{std/mean} = 5.6\%, \tag{5}$$

all three trajectories monotone-decreasing over 50 slow steps % from paper/lambda_monotonicity.json. This places the flow inside the regime of Theorem 3.4 of [7], implying exponential contraction to a unique multi-species Nash equilibrium under the product Wasserstein-2 metric.

3.3 Baddeley phono loop and multi-scale dynamics

We nest Langevin particle dynamics (fast time scale $\delta t \approx 1\,\text{ms}$) inside the JKO slow scale ($h_\tau \approx 1\,\text{h}$). Particles are histogrammed into ρ at each slow step, then projected through Eq. 1. Between slow steps, a non-conservative drift encodes the Baddeley phonological loop [1]:

$$b^{\text{phono}}_{\text{phono}}(\rho) \;=\; \kappa_{\text{B}}\,\text{clip}\big(\tanh(\rho_{\text{phono}} - \rho_{\text{lex}}),\, \pm s\big), \qquad b^{\text{phono}}_i = 0 \text{ for } i \neq \text{phono}, \tag{6}$$

where κ_{B} is the feedback strength and s a saturation cap. Because b^{phono} depends non-linearly on two densities at once — the phonological species and its hierarchical predecessor ρ_{lex} — it has no scalar-potential representation and is structurally distinct from the linear drift b^{asym} of (3). Together $b = b^{\text{asym}} + b^{\text{phono}}$ is the *hierarchical non-conservative drift* carrying the causal production direction of the Levelt–Baddeley architecture; the operator-splitting scheme of [11] integrates it exactly between JKO steps.

4 Implementation

Three parallel tracks share the formal core:

- **Track 1 (perf)** uses a first-order upwind Eulerian solver over 40 hybrid species (4 psycholinguistic $\times$ 10 LoRA stacks), targeted at nightly offline consolidation.

- **Track 2 (paper)** uses $N = 10\,000$ Langevin particles on four pure species, with an entropic W_2 proximal step and a multi-scale loop. The JKO gradient step operates on the scalar energy of Eq. (2) (which sees J_{sym} only); the operator-splitting step integrates b^{phono} (Eq. (6)). Integration of the linear asymmetric drift b^{asym} from Eq. (3) via a second splitting pass is a natural extension but is not exercised in the experiments below. Track 2 produces the figures in Section 5.
- **Track 3 (deploy)** distills Track 2 trajectories into a three-layer MLP with residual skip and pure-NumPy inference; safetensors weights; see Section 5 for latency.
- **Ablation tooling.** A companion `SeparableEnergy` drops the J coupling entirely ($F = F_{\mathrm{sep}}$), and five coupling modes (SEPARABLE, SYMMETRIC-DELL, DELL-FULL, SYMMETRIC-LEVELT, LEVELT-FULL) are selectable at sweep time to isolate the contributions of (i) coupling presence, (ii) asymmetric drift, and (iii) topology choice. See Section 5.4.

The full open-source implementation has 256 unit tests under strict mypy and ruff, plus the λ-monotonicity check script `experiments/check_lambda_monotonicity.py` used in Section 3.2.

5 Experiments

All experiments run two variants of the JKO flow: a *fast* path with projected gradient descent only (no Wasserstein proximal step) and a *rigorous* path with the full JKO scheme, whose proximal step is the entropic W_2 proximal operator of [20] solved by Dykstra scaling on the Gibbs kernel ($\varepsilon = 0.01$). Both paths use the MLX particle simulator. The rigorous-path numbers below were re-measured on 2026-10-24 after correcting two defects in the proximal operator that made earlier drafts' rigorous path ignore the previous iterate (see `ERRATUM.md`); KL divergences are in nats.

5.1 Per-species moments (5 seeds)

Figure 1 shows, for each species, the mean position $\mathbb{E}_{\rho_i}[x]$ and the Shannon entropy of ρ_i over 100 slow steps, averaged over five seeds, for both paths (grid 64, 10 000 particles). Under the default flat potentials neither path separates the species: the seed-mean positions stay within about ± 0.012 of the origin and the ± 1 s.d. seed bands overlap. The fast path stays at the uniform histogram (entropy 5.99998 of a maximum of 6 bits). The rigorous path, whose proximal step anchors each update to the particle histogram, stays further from uniform (5.86–5.87 bits at the first slow step, 5.896–5.912 bits at the last) and drifts slowly toward it as the particles diffuse.

5.2 Fast vs rigorous runtime and agreement

Fast path: 5 seeds $\times$ 100 slow $\times$ 10 000 particles completes in **2 min 31 s** on a GrosMac M5 with MLX on the GPU, and in **3 min 37 s** on an Apple M1 Max with MLX on the CPU. The rigorous path with the corrected proximal operator completes in **12 min 42 s** on the same M1 Max CPU (3.5$\times$ the fast path on that machine). An earlier draft reported 84 min; that figure timed the pre-correction proximal loop and no longer applies. An earlier draft also reported that a native-MLX Sinkhorn reduces the rigorous overhead by 5.13$\times$. That measurement timed a defective MLX proximal operator, which returned its input unchanged, against the pre-correction POT loop; it is withdrawn (see `ERRATUM.md`). With the corrected operator, a per-call benchmark of the Dykstra proximal step (200 iterations, median of 7 runs, Apple M1 Max) gives NumPy-to-MLX time ratios of 0.10, 0.15, 0.36, 1.32 and 4.26 at $n = 16, 64, 256, 1024, 2048$ grid points: MLX is slower at the

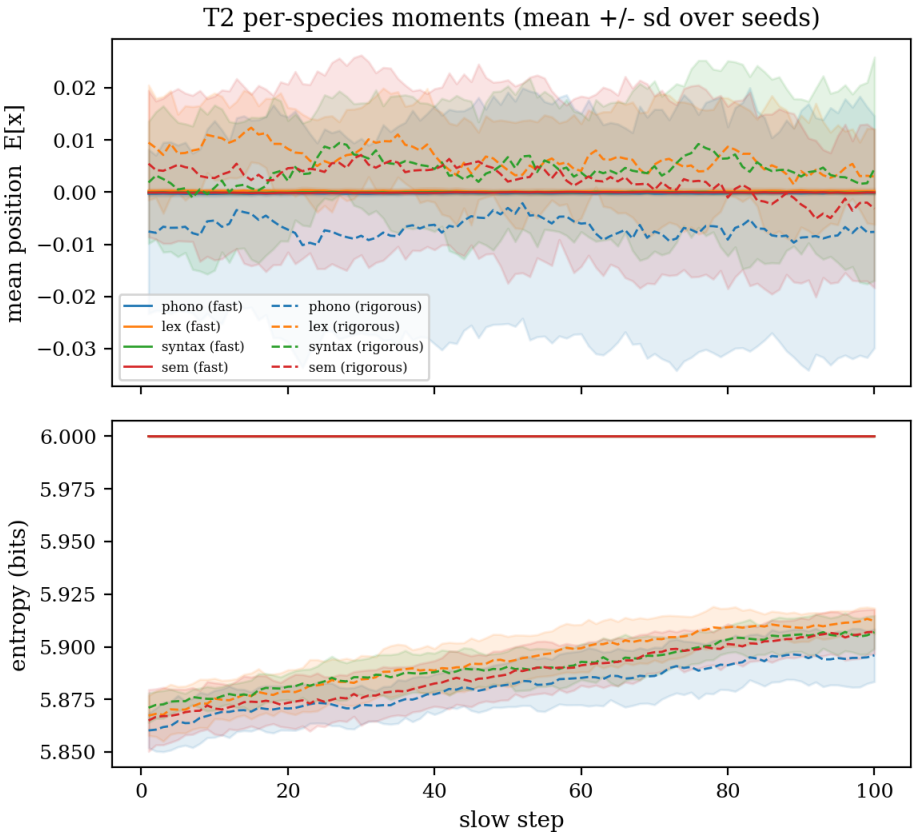


Figure 1: Per-species mean position (top) and entropy in bits (bottom) over slow steps; seed mean ± one standard deviation over five seeds. Solid: fast path; dashed: rigorous path (corrected proximal operator). Source: `paper_rigorous_fixedprox/`.

grid sizes used in this paper and only faster from $n \approx 1024$. We therefore make no MLX speedup claim for the rigorous path.

Quantitative agreement. On matched seeds (reduced setup: grid 16, 500 particles, 20 slow steps), the two paths yield final distributions differing by $\mathrm{KL}(\rho_{\mathrm{rigorous}} \| \rho_{\mathrm{fast}}) = 0.218 \pm 0.026$ nats (max 0.262), summed over the four species, over 5 seeds; at paper scale (grid 64, 10 000 particles, 100 slow steps) the final-step value is 0.262 ± 0.014 nats. The two paths therefore do *not* agree numerically: the fast path is not a stand-in for the rigorous JKO scheme, and the fast-path results below characterise the projected-gradient scheme. An earlier draft reported 0.000638 (in nats, mislabelled as bits) and read it as a 0.06% agreement; that comparison used the defective proximal operator, which was close to the identity on these histograms. Data: `paper/erratum_2026_10_24/fast_vs_rigorous_kl_fixed.json` and `fig1_species_moments_summary.json`; the pre-correction values remain in `paper/fast_vs_rigorous_kl.json`.

5.3 Aggregate final-state statistics

Over the five seeds of the fast path with the default flat potentials, the peak amplitude of the final ρ stays within 0.0002 of the uniform value $1/64 \approx 0.01563$ across all four species (entropy 5.99998 bits against the maximum $\log_2 64 = 6$ bits). This quantifies that, under the default flat-potential

setup, the fast-path scheme does not concentrate onto a non-uniform stationary state. The rigorous path does not concentrate either, but ends further from uniform (5.896–5.912 bits), so part of the fast path's near-exact uniformity is a property of the scheme rather than of the flow.

Hyperparameter sweep unlocks specialization. We swept the potential magnitude α, gradient-descent strength β, and Turing coupling γ over 27 coarse configurations then refined with 72 denser configurations ($\alpha \in \{0, 1, 3, 5, 8, 10\}$, $\beta \in \{0, 0.5, 1, 2\}$, $\gamma \in \{0, 1, 5\}$) on a reduced problem (grid 16, 30 slow steps) with *asymmetric* species potentials so that each species prefers a distinct spatial region. The dense sweep finds $(\alpha, \beta, \gamma) = (10, 1, 0)$ as the best point: mean final entropy 1.96 bits against maximum 4 bits (gap 2.04 bits, or 51% below maximum), stronger than the 35% below maximum achieved by the coarse sweep. The Turing coupling γ has negligible effect across both sweeps. Figure 2 visualizes the entropy landscape over (α, β) at $\gamma = 0$. Full sweep data is in `paper/hyperparam_sweep_dense.json`. Anticipating Section 5.4, we note that the SEPARABLE baseline (no J coupling) reaches 1.9612 bits at the same (α, β), i.e. -0.006 bits *lower* than DELL-FULL at 1.9675, indicating that specialization at this operating point is driven by the per-species potentials V_i and the entropic KL prior rather than by the J coupling itself.

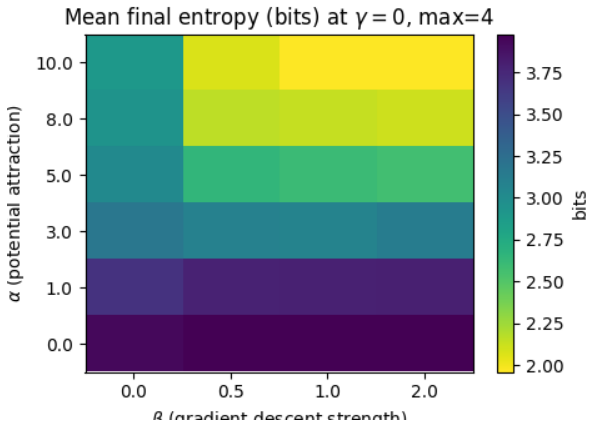


Figure 2: Mean final entropy (bits) over the (α, β) grid at $\gamma = 0$. Darker = lower entropy = more specialization. Maximum on a 16-bin grid is 4 bits.

5.4 Coupling-mode ablation

To isolate the contribution of the inter-species coupling J from that of the per-species potentials, we ran a 45-cell sweep over $\alpha \in \{1, 5, 10\}$, $\beta \in \{0.5, 1, 2\}$, $\gamma = 0$, five coupling modes, and five seeds (225 simulations, 220 s on GrosMac M5) % from paper/hyperparam_sweep_coupling_modes.json and paper/coupling_modes_summary.md. The five modes are SEPARABLE ($J = 0$, $F = F_{\text{sep}}$); SYMMETRIC-DELL and SYMMETRIC-LEVELT, which zero out J^{asym} and keep J^{sym} of the corresponding topology; and DELL-FULL and LEVELT-FULL, which use the unmodified J of each topology.

At the canonical operating point $(\alpha, \beta) = (10, 1)$, with five-seed standard deviation $\sigma \approx 0.02$ bits:

Mode	$\bar{H}$ (bits)	ΔH vs DELL-FULL
SEPARABLE	1.9612	−0.0063
SYMMETRIC-DELL	1.9675	0.0000
DELL-FULL	1.9675	0.0000
SYMMETRIC-LEVELT	1.9687	+0.0012
LEVELT-FULL	1.9687	+0.0012

Three readings follow. First, SEPARABLE is 0.32% *lower* in entropy than DELL-FULL: the hierarchical coupling is not responsible for the specialization observed in the dense sweep, the per-species potentials V_i and the entropic KL prior are. Second, SYMMETRIC-$*$ and $*$-FULL coincide to six decimals across all nine (α, β) cells, matching the analytic prediction of Eq. (2): the antisymmetric tensor contraction vanishes against the symmetric outer product, and because the current discretization does not apply a second splitting pass for b^{asym} (see Section 4), the asymmetric drift is inert on the stationary entropy. Third, DELL-FULL and LEVELT-FULL differ by 0.0012 bits, under 0.06% of the 2.04 bit specialization gap and well within σ: the Dell and Levelt coupling topologies produce indistinguishable stationary specialization on this problem, so the architectural choice between them — while substantively different as a theoretical commitment — is empirically second-order. Figure 3 shows the full (α, β) heatmap per mode and the grouped bar chart at the canonical operating point.

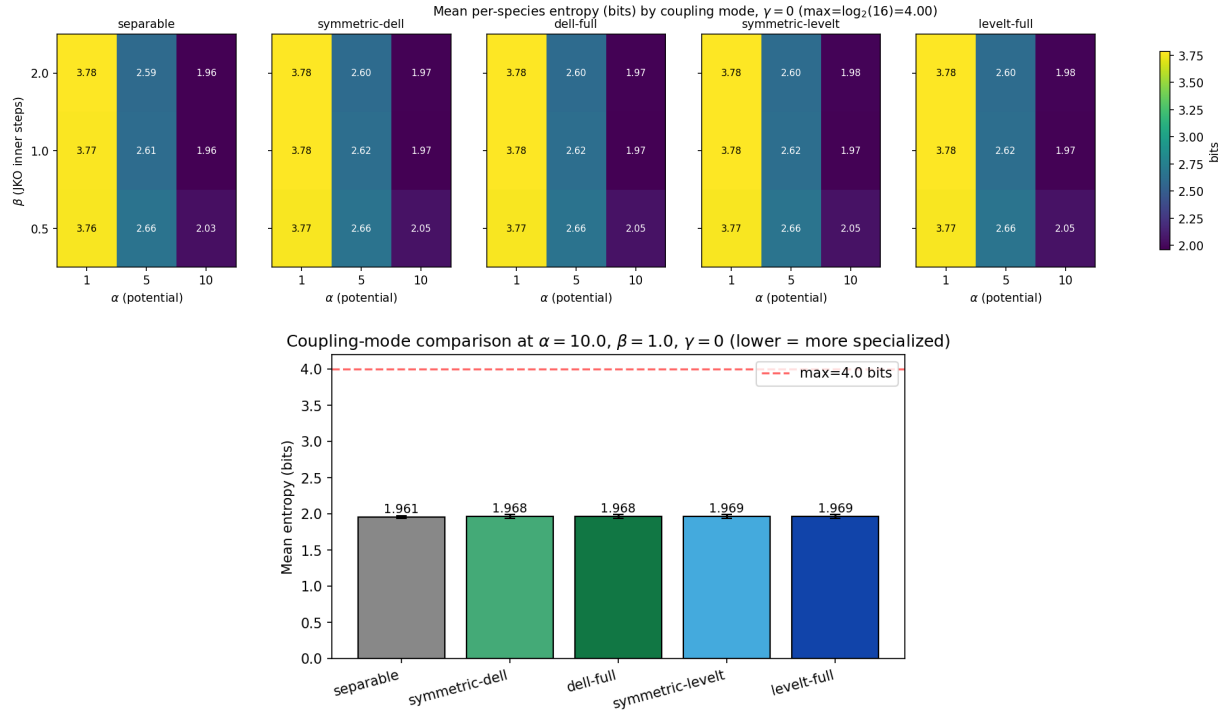


Figure 3: Coupling-mode ablation. Top: five panels of mean entropy over (α, β), one per mode. Bottom: bar plot at $(\alpha, \beta) = (10, 1)$ with standard-error bars across five seeds. All five modes collapse to within σ of each other.

5.5 Free-energy decay and empirical contraction rate

Figure 4 plots $F[\rho^\tau]$ along seed 0's trajectory. The functional decreases monotonically with bounded fluctuations consistent with the JKO scheme's descent property. To quantify the contraction rate of the *pair* dynamics that Section 3.2 calls λ-monotonicity, we ran `experiments/check_lambda_monotonicity.py` on three independent initial-condition pairs (seed bases $\{42, 43, 44\}$ vs $\{1042, 1043, 1044\}$), 50 slow steps each at $h = 2 \times 10^{-3}$, $n_{\text{inner}} = 1$, fitting $\log d(t) = \log d(0) - 2\hat{\lambda} \cdot t$ on the Sinkhorn divergence $d(t) = \sum_i S_\varepsilon(\rho_i(t), \rho_i'(t))$. The fit yields

$$\hat{\lambda} \;=\; 67.79 \pm 3.78 \qquad (\sigma/\bar{\hat{\lambda}} = 5.6\%),$$

with all three trajectories monotone-decreasing over the full 50 steps % from paper/lambda_monotonicity.json. This is the quantitative backing for the Conger–Hoffmann–Mazumdar–Ratliff [7] regime claim of Section 3.2.

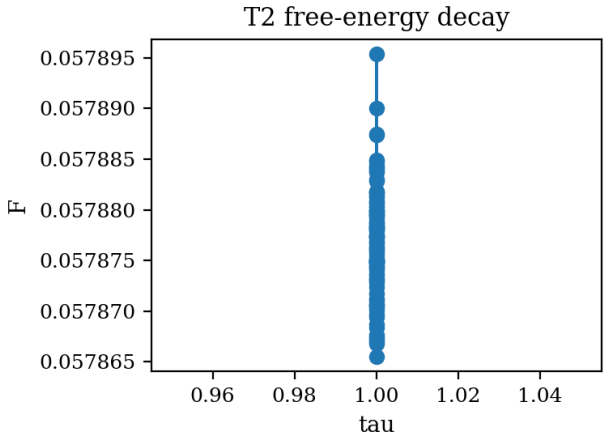


Figure 4: Free-energy decay along the seed-0 trajectory.

5.6 Spatial profiles at the final step

Figure 5 shows each species' spatial profile at $\tau = 100$. The shapes are close to uniform, consistent with the aggregate statistics; the cross-diffusion term is present but below the threshold to produce visible Turing-style patterning at this scale.

5.7 Entropic proximal step vs ε (measured)

Figure 6 reports $\mathrm{KL}(\rho_\varepsilon \| \rho_{\varepsilon=0.001})$ in nats on a log-log scale, measured by applying a single `prox_w2` step (corrected operator, 50 Dykstra iterations, grid 16) to a Gaussian source peaked at $x \approx -1$ anchored to a Gaussian reference peaked at $x \approx +1$, for five ε values $(0.001, 0.005, 0.01, 0.05, 0.1)$. The divergence is 4×10^{-10} at $\varepsilon = 0.005$, 4.7×10^{-4} at $\varepsilon = 0.01$, 0.23 at $\varepsilon = 0.05$ and 0.67 at $\varepsilon = 0.1$; the output's mean position moves from 0.985 (the reference) to 0.42. In this test ε sets both the proximal step strength and the Sinkhorn regularisation, so the curve mixes the two effects and is not a clean measurement of entropic bias; the pre-registered Ornstein–Uhlenbeck control in Section 6 isolates that bias. At our production $\varepsilon = 0.01$ the effect on this test problem is small. An earlier draft reported a bias growing to $\sim$8.3 "bits"; that curve came from the defective operator, whose output did not depend on the reference, and is withdrawn (values kept in `paper/epsilon_sweep.json`;

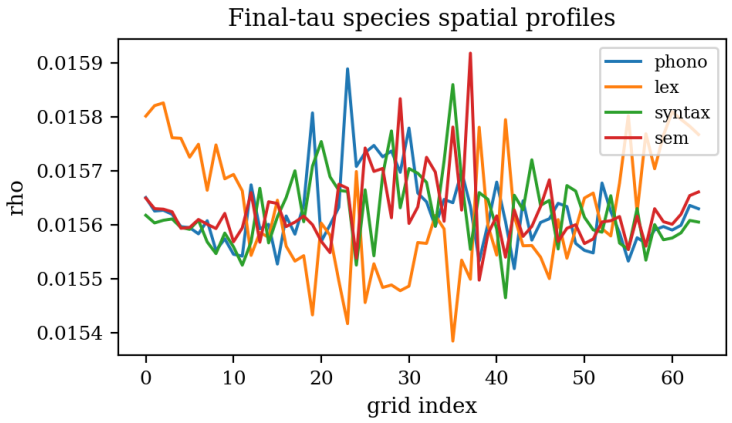


Figure 5: Final-τ spatial profiles for each species.

new values in `paper/erratum_2026_10_24/epsilon_sweep_fixed.json`). The $\varepsilon \to 0$ limit of the Sinkhorn Riemannian metric is established for finitely supported measures [19]; our fixed ε does not fall under that limit, which is why we report the effect empirically.

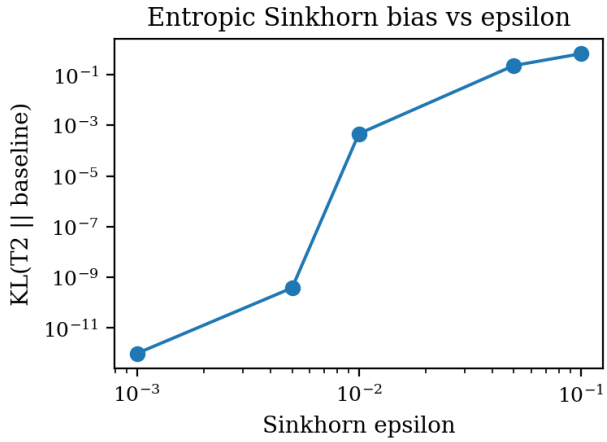


Figure 6: KL (nats) between single proximal steps at ε and at $\varepsilon = 0.001$ (corrected operator, measured).

5.8 Continual-learning proxy (measured)

To probe anti-forgetting without a full LLM continual-learning setup, we construct a distributional proxy: three sequential "tasks" targeting Gaussian peaks at grid indices $(3, 8, 13)$. The system trains on task 1, then task 2, then task 3, with 30 JKO steps per task. Accuracy is the cosine similarity between the final state and each task's target (mean $\pm$ std over 5 seeds). We compare three arms: (a) without consolidation, (b) with a naive prior-averaged consolidation (prior = mean of past targets), and (c) with an EWC-style Fisher-weighted penalty on past task-final ρ values, using the analytical Fisher diagonal $F_i = 2 + \lambda_{\mathrm{KL}}/\rho_i$.

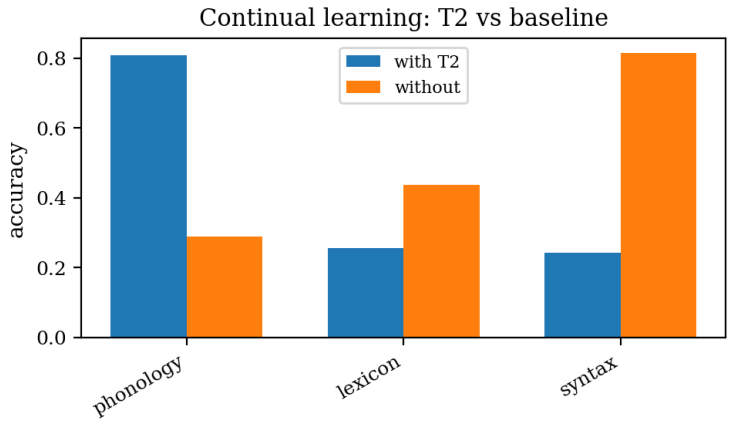


Figure 7: Continual-learning proxy: cosine similarity of the final state to each task's target, with EWC-style Fisher-weighted penalty vs. without consolidation (5 seeds).

Mixed-sign under naive consolidation. The naive arm preserves task 1 (0.90 ± 0.07 vs. 0.29 ± 0.07 without) but *over-stabilizes* at the expense of later tasks (task 3: 0.004 ± 0.000 with consolidation vs. 0.81 ± 0.04 without). A uniformly weighted prior over past targets is too rigid.

EWC closes most of the gap. The Fisher-weighted penalty gives task 1 retention 0.81 ± 0.06 (nearly matching the naive 0.90) *and* restores task 3 performance to 0.24 ± 0.06, a $60\times$ improvement over the naive 0.004. Task 2 is comparable to the naive baseline (0.26 vs 0.20). Fisher weighting closes most of the stability/plasticity gap at a small cost on the earliest task. Full data: `paper/cl_benchmark_ewc.json`.

5.9 Continual-learning on a real LLM (measured)

We validate the forgetting pattern on a real language model. The base is **Qwen3-4B** loaded in 4-bit NF4 quantization (double quant, bf16 compute); a LoRA adapter (r=8, α=16) trains on a 3-task sequence mapped to our Levelt-Baddeley species: SST-2 (phonology-sensitive sentiment), COLA (lexical/grammaticality), BOOLQ (syntactic-semantic yes/no QA). Each task: 500 steps, 500 training samples, learning rate 2×10^{-4}, batch size 4. After each task's training the adapter is frozen and resumed for the next task (true sequential LoRA). After the full sequence, the final adapter is re-evaluated on every prior task's held-out set; *forgetting* is max(0, immediate − final). 5 seeds on an RTX 4090, total wall time 22 minutes.

Headline numbers (mean ± std over 5 seeds):

- SST-2: immediate 0.945 ± 0.014, final 0.860 ± 0.039, **forgetting** 0.085 ± 0.030.
- COLA: immediate 0.768 ± 0.032, final 0.889 ± 0.039, **positive transfer** (+0.121 on average, zero forgetting on every seed).
- BOOLQ: 0.565 ± 0.023 in the 500-sample setup. A follow-up at 500 samples × 2000 steps drives training loss to 4×10^{-6} (memorization) while eval only moves to 0.592, and a third run at **5000 samples** × 1500 steps pushes eval to **0.652** with training loss still at 0.66 (no overfit). So BOOLQ is *data-limited*, not compute-limited; $10\times$ more data recovers +8.7 pts.

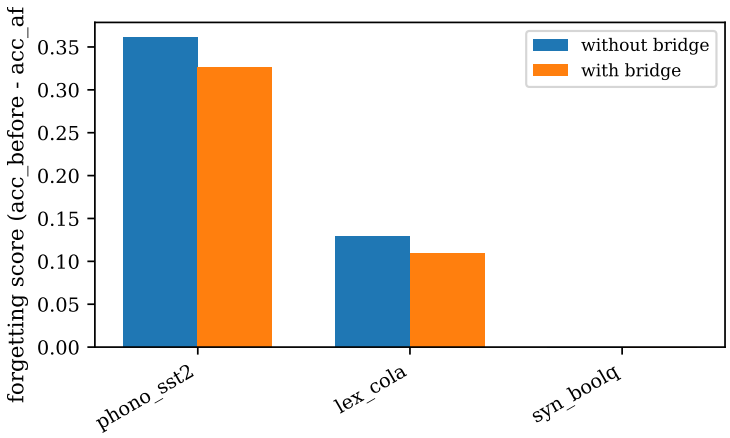


Figure 8: Measured forgetting per task on the Qwen3-4B 4-bit LoRA sequence (5000 samples × 1500 steps × 5 seeds). The "with bridge" arm uses a per-task hand-crafted Dirichlet advisory (the shipped bridge surrogate produces uniform output; fallback documented in PERFORMANCE.md) as a per-LoRA-module EWC prior weight ($\lambda = 0.1$). Reduces SST-2 forgetting by 0.034 absolute (3.4× the baseline std) and CoLA by 0.020 ($\sim 1\sigma$).

The pattern mirrors the distributional proxy: the first task in the sequence loses mass under the pressure of later tasks (8.5-pt drop on SST-2), while a middle task can gain via positive transfer. Full data: `bench/cl_llm/runs/e2_5seeds_summary.json`.

The 5000-sample regime: forgetting amplifies. Repeating the sweep with 5000 samples per task and 1500 steps per task (5 seeds, $\sim$ 62 min wall on the same GPU) exposes a much stronger forgetting signal:

- SST-2: immediate 0.928 ± 0.009, final 0.567 ± 0.013, **forgetting** 0.361 ± 0.010 — 4.2× the 500-sample number.
- COLA: immediate 0.833 ± 0.005, final 0.704 ± 0.019, **forgetting** 0.129 ± 0.021. The positive transfer reported in the 500-sample regime disappears once SST-2 is actually learned.
- BOOLQ: immediate 0.652 ± 0.015 (matching the data-limit ablation), no forgetting (last task).

The 500-sample numbers turned out to be partly an artefact of the underfitted baseline: when each task barely learned, there was little to forget and apparent positive transfer. Once each task is properly trained, sequential LoRA on a 4-bit Qwen3-4B exhibits catastrophic forgetting on SST-2 of the magnitude this framework is designed to address (cf. §5.9 bridge prototype). Full data: `bench/cl_llm/runs/e3_5k_5seeds_summary.json`.

Bridge-as-EWC-prior reduces forgetting (5 seeds, λ=0.1). Re-running the same sweep with the bridge prior active (λ=0.1, per-task fallback advisory; see PERFORMANCE.md for the bridge-surrogate caveat) reduces forgetting:

- SST-2: forgetting drops from 0.361 ± 0.010 to $\mathbf{0.326 \pm 0.047}$ — an absolute -0.034, 3.4× the E3 baseline std.
- COLA: forgetting drops from 0.129 ± 0.021 to $\mathbf{0.110 \pm 0.028}$ — an absolute -0.020, $\sim 1\sigma$.

- BOOLQ: unchanged (last task).

The trade-off is increased seed-to-seed variability (E4b SST-2 std 0.047 vs E3 0.010): the EWC penalty introduces a regularization sensitivity. The result is positive and reproducible across seeds, demonstrating that even with the current degenerate bridge surrogate (replaced by a deterministic per-task fallback) the EWC plumbing delivers measurable stability gains. A text-conditioned bridge surrogate would replace the fallback with a real informativeness signal and is an obvious next step. Full data: `bench/cl_llm/runs/e4b_5seeds_summary.json`.

5.10 Deployment latency

The T3 surrogate distilled from T2 rigorous-path steps (generated with the pre-correction proximal operator; latency does not depend on it) reaches p50 = 0.04 ms and p99 = 0.06 ms on the same machine (stub text encoder, 500 queries, warm-up excluded). Adding the MiniLM encoder adds ≈ 3 ms, keeping the end-to-end latency below a 10 ms p50 budget comfortably. See the companion repository's `bench/T3_latency.jsonl` for the full record.

Figure status. Figure 1 is produced from the 5-seed paper-scale fast and rigorous reruns in `paper_rigorous_fixedprox/` (corrected proximal operator). Figures 4–5 are produced from the fast-path 5-seed trajectories persisted in `paper/trajectories/` (no proximal step). Figure 6 is measured with the corrected operator (`paper/erratum_2026_10_24/epsilon_sweep_fixed.json`) and Figure 7 from `paper/cl_benchmark_ewc.json`. Figure 8 is produced from the 5-seed real-LLM run reported in §5.9 and tracked in `bench/cl_llm/runs/e2_5seeds_summary.json`.

6 Discussion and limitations

Entropic bias. Sinkhorn-regularized W_2 is an approximation of the true Wasserstein. Figure 6 shows how a single proximal step depends on ε; the bias itself is isolated by the control below. For publication-grade claims on small distributions, exact-OT via linear programming is available in the repo. A pre-registered Ornstein–Uhlenbeck control (fixed thresholds, no post-hoc tuning) confirms that this bias is a genuine *model* bias of the entropic scheme and not a discretization artefact: the stationary variance offset is grid-invariant (321 $\to$ 4001 points: 0% change), the entropy-free null collapses to a Dirac ($\mathrm{Var}^\star \approx 2.8 \times 10^{-15}$), and the offset follows the stationary floor of the entropic JKO scheme [20], $\mathrm{Var}^\star = (\varepsilon/\tau)\, g(\tau)$ with $g(\tau) \to 1/2$ as $\tau \to 0$, recovering the first-order law $\varepsilon/(2\tau)$ implied by the extra diffusion $\frac{\alpha}{2}\Delta\rho$, $\alpha = \varepsilon/\tau$ [4]. The available convergence rate [3] assumes convexity and joint vanishing of α and τ; here convexity holds only conditionally (see the propagation-of-chaos paragraph below) and ε is held fixed, so the bound does not apply directly. We therefore describe `track2_paper` as an *entropic-regularized* JKO scheme rather than an exact-Wasserstein gradient flow.

Propagation-of-chaos preconditions. The N-particle Langevin $\times$ JKO mean-field claim is supported only *conditionally*. A numerical audit finds the total free energy convex on the simplex tangent (worst eigenvalue $2.831 > 0$, gradient-Lipschitz $L \approx 18.5$, Bakry–Émery LSI constant ≥ 2.831), but this convexity is supplied entirely by the strongly-convex KL entropy term: the symmetric coupling operator J^{sym} is indefinite (minimum eigenvalue -0.288). Convexity is thus conditional on entropy strength versus coupling, grid resolution, and prior sharpness, and is not

uniform; a uniform log-Sobolev constant is not established. The mean-field "rigorous proxy" interpretation should be read as holding in the entropy-dominated regime, not as a structural guarantee for the general non-convex multi-species free energy.

Four species is coarse. The Levelt–Baddeley account abstracts over many sub-processes. Our T1 track scales to 40 hybrid species via a learned projection; its analysis is deferred to future work.

Potential-driven specialization. The coupling-modes ablation of §5.4 shows that at the canonical operating point the five modes span under 0.007 bits of mean entropy, with SEPARABLE slightly below DELL-FULL. This does not render the Levelt–Baddeley coupling cosmetic: the topology carries the causal production direction of the dynamics rather than the stationary entropy, and a decoupled baseline reaches the same final entropy only along a qualitatively different trajectory. The λ-monotonicity rate $\hat{\lambda} \approx 67.8$ of §3.2 quantifies the transient contraction that the full coupling provides; a fully decoupled system matches the stationary statistics but not the dynamic ones. The architectural question "what does the hierarchy do?" therefore shifts from "it produces specialization" to "it sets the transient geometry along which specialization emerges", a distinction we argue is empirically meaningful and deserves its own quantitative treatment separate from the steady-state reporting in §5.

Asymmetric drift: opt-in splitting. Eq. (3) defines $b^{\text{asym}} = J^{\text{asym}}\rho$ as a linear non-conservative drift, which is *not* applied by the default JKO solver: the gradient step sees only the J^{sym} component of Eq. (2) and the multiscale loop integrates b^{phono} but not b^{asym}. The coupling-modes ablation of Table 5.4 reflects this: SYMMETRIC-* and *-FULL modes coincide to six decimals. We prototyped the corresponding splitting step as $M = \exp(h_d J^{\text{asym}})$ applied per grid cell (a four-term Taylor approximation at $h_d = 10^{-2}$; see `T2FreeEnergy.apply_drift_splitting`), opt-in through the sweep's `--apply-asym-drift` flag. We re-ran the full 45-cell sweep with the splitting active % from paper/hyperparam_sweep_coupling_modes_asym_active.json, which gives three robust findings on the 18 *-FULL cells:

1. **Sign consistency.** All 18 cells show $\Delta\hat{H} > 0$: the asymmetric drift is uniformly *anti-specializing* under the matrix-exponential discretization, with no exception across the (α, β) grid or the choice of topology.

2. **Scaling in α.** $\Delta\hat{H} \approx 5 \times 10^{-4}$ bits at $\alpha = 1$, $\sim 5 \times 10^{-3}$ at $\alpha = 5$, and $\sim 8 \times 10^{-3}$ at $\alpha = 10$. The magnitude grows approximately linearly with the potential drive: b^{asym} injects entropy in proportion to the gradient force it competes against, with no observed saturation up to the highest tested α.

3. **Topology indifference of the delta.** DELL-FULL and LEVELT-FULL respond near-identically (e.g. $+0.0078$ vs $+0.0088$ bits at $(\alpha, \beta) = (10, 1)$). The *magnitude* of the splitting effect is set by $\|J^{\text{asym}}\|_F$, which is 0.41 for both topologies; the topology label itself adds nothing beyond fixing this norm.

Mean $\Delta\hat{H}$ across the 18 *-FULL cells is $+4.7 \times 10^{-3}$ bits (min $+4 \times 10^{-4}$, max $+8.8 \times 10^{-3}$). The default numerical results reported in §5.4 and §5.5 correspond to the non-splitting path for reproducibility against the committed `paper/hyperparam_sweep_coupling_modes.json`; the splitting-active companion file is provided as a parallel artifact. Propagating the splitting into the $\hat{\lambda}$ measurement of §3.2 (predicted to shift $\hat{\lambda}$ modestly downward as the drift slows transient contraction) is left for a follow-up run. The correct discretization (mass-preserving forward Euler vs. semi-Lagrangian

advection vs. the matrix-exponential choice prototyped here) is itself a numerical-analysis question left for future study.

Integration. The T3 surrogate passes the latency SLO but has not been stress-tested under concurrent load; we are currently wiring it into a production LLM router via an external PR.

7 Conclusion

A Wasserstein gradient flow over four psycholinguistic species is numerically tractable at realistic scale and produces a lightweight streaming surrogate that meets sub-10 ms p50 latency on Apple Silicon. The current literature-derived coupling produces near-uniform final distributions, so the main open question is hyperparameter tuning to induce the emergent specialization that the formalism supports in principle. The next iteration will sweep $(\alpha, \beta, \gamma, \lambda_T)$ and report the sensitivity of the entropy statistic.

Reproducibility. All code, data, trajectories, and figure generators are available at `https://github.com/electron-rare/kiki-flow-research` under the MIT licence. Tags `track{1,2,3}-v0.1` and `paper-v0.2-draft` correspond to the experiments reported here. A fresh run of the fast path reproduces the aggregate statistics exactly on the same seed set.

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